

**Remark:** The following definitions (or results) are already proved in class or on previous ICA, Wbwk, Hmwk assignments, and in Sections 4.4, 4.5, and 4.6 of the textbook.

**§ Set Equality** Let  $A$  and  $B$  be sets. The sets  $A$  and  $B$  are equal ( $A = B$ ) iff  $A \subseteq B$  and  $B \subseteq A$ .

**§ Subset Relation** Let  $A$  and  $B$  be sets.  $A$  is a subset of  $B$  ( $A \subseteq B$ ) iff for all  $x \in A$  only if  $x \in B$ .

**§ Union of Sets** Let  $A$  and  $B$  be sets. The union of  $A$  and  $B$ , denoted  $A \cup B$ , is the set of all elements that are in  $A$ , or in  $B$ , or in both. That is,  $A \cup B = \{x : x \in A \text{ or } x \in B\}$ .

**§ Intersection of Sets** Let  $A$  and  $B$  be sets. The intersection of  $A$  and  $B$ , denoted  $A \cap B$ , is the set of all elements that are in both  $A$  and  $B$ . That is,  $A \cap B = \{x : x \in A \text{ and } x \in B\}$ .

**§ Set Difference** Let  $A, B$  be subsets of a universal set  $\mathcal{U}$ . The set difference  $A \setminus B$  is defined to be the set  $\{x \in \mathcal{U} \mid x \in A \text{ and } x \notin B\}$ .

*It is sometimes called the relative complement of  $B$  in  $A$ , and denoted in the textbook as  $A - B$ .*

**§ Power Set** Let  $A$  be a set. The power set of  $A$ , denoted  $\mathcal{P}(A)$  or  $2^{|A|}$ , is the set of all subsets of  $A$ . That is,  $\mathcal{P}(A) = \{S \in \mathcal{U} : S \subseteq A\}$ .

**§ Complementary Set** Let  $A$  be a subset of the universal set  $\mathcal{U}$ . The complement of  $A$ , denoted  $A^c$  or  $\overline{A}$ , is the set of all elements in  $\mathcal{U}$  that are not in  $A$ . That is,  $\overline{A} = \{x \in \mathcal{U} : x \notin A\}$ .

**§ Cartesian Product** Let  $A$  and  $B$  be sets. The Cartesian product of  $A$  and  $B$ , denoted  $A \times B$ , is the set consisting of all ordered pairs:

$$A \times B = \{(a, b) \mid a \in A, b \in B\}.$$

**§ Pairwise Disjoint Sets** Let  $A_1, A_2, \dots, A_n$  be sets. These sets are **pairwise disjoint** if for all  $i \neq j$ , we have  $A_i \cap A_j = \emptyset$ .

**§ Function** Let  $A$  and  $B$  be nonempty sets. A **function**  $f$  from  $A$  to  $B$ , denoted  $f : A \rightarrow B$ , is a relation,  $f \subseteq A \times B$  such that

1. For every  $a \in A$ , there exists a  $b \in B$  such that  $f(a) = b$ .
2. For every  $a \in A$  and for all  $b, c \in B$ , if  $f(a) = b$  and  $f(a) = c$ , then  $b = c$ .
3. The set  $A$  is called the **domain** of  $f$ , and the set  $B$  is called the **codomain** of  $f$ .
4. If  $f(a) = b$ , then  $b$  is called the **image** of  $a$  under  $f$ .
5. The function maps  $a$  to  $b$ , and  $f$  is often called a mapping.

**§ Domain and Codomain** If  $f : A \rightarrow B$  is a function, then  $A$  is called the **domain** of  $f$ , and  $B$  is called the **codomain** of  $f$ .

**§ Image and Inverse Image** Let  $f : A \rightarrow B$  be a function,  $C \subseteq A$ , and  $D \subseteq B$ . The **image**  $f(C)$  is the set  $\{f(c) \in B : c \in C\} \subseteq B$ . The **inverse image**  $f^{-1}(D)$  is the set  $\{a \in A : f(a) \in D\} \subseteq A$ .

**§ Injective, Surjective, Bijective Functions** Let  $f : A \rightarrow B$  be a function.

- $f$  is **injective** (one-to-one) iff for all  $a_1, a_2 \in A$ , if  $f(a_1) = f(a_2)$ , then  $a_1 = a_2$ .
- $f$  is **surjective** (onto) iff for every  $b \in B$ , there exists  $a \in A$  such that  $f(a) = b$ .
- $f$  is **bijective** iff it is both injective and surjective.

**§ Theorem** The inverse relation  $f^{-1}$  is a function if and only if  $f$  is both injective and surjective.

**Proof:** We consider the inverse relation  $f^{-1} \subseteq B \times A$  for the function

$$f : A \rightarrow B \quad \text{on nonempty sets } A, B.$$

Assume  $f$  is a bijection, i.e., both one-to-one and onto. Let  $b \in B$ . Since  $f$  is onto, by definition, there exists  $a \in A$  with  $f(a) = b$ . So  $(a, b) \in f$  and  $(b, a) \in f^{-1}$ . Hence for all  $b \in B$  there exists  $a \in A$  such that  $(b, a) \in f^{-1}$ .

Now let  $a, a' \in A$ . Suppose  $(b, a) \in f^{-1}$  and  $(b, a') \in f^{-1}$ . Then  $(a, b), (a', b) \in f$ , so  $f(a) = b$  and  $f(a') = b$ . But  $f$  is one-to-one by assumption, and so  $a = a'$ . Hence we have shown that each  $b \in B$  has at least one image under the inverse relation  $f^{-1}$  and no more than one. Thus the inverse relation is a function whenever  $f$  is a bijection.

Conversely, ...



1. Let  $A$ ,  $B$ ,  $C$ , and  $D$  be subsets of a universal set  $\mathcal{U}$ . Prove the following propositions.<sup>1</sup>

*This proves that the three subsets are a partition of  $(A \cup B)$ . Partitions will be very important to understanding equivalence relations.*

(a)  $(A \cup B) = (A \cap B) \cup (A \setminus B) \cup (B \setminus A)$ .<sup>2</sup>

**Proof:** By definition of Set Equality,  $(A \cup B) = (A \cap B) \cup (A \setminus B) \cup (B \setminus A)$  requires showing both directions of Subset Relations, as follows.

$(\subseteq)$  Let  $x \in A \cup B$  be arbitrary. Then,  $x \in A$  or  $x \in B$  by definition of Union of Sets. Consider the following cases.

- If  $x \in A$  and  $x \in B$ , then  $x \in A \cap B$  by definition of Intersection of Sets.
- If  $x \in A$  and  $x \notin B$ , then  $x \in A \setminus B$  by definition of Set Difference.
- If  $x \notin A$  and  $x \in B$ , then  $x \in B \setminus A$  by definition.

In all cases,  $x \in (A \cap B) \cup (A \setminus B) \cup (B \setminus A)$ . Hence, proof by cases,  $(A \cup B) \subseteq (A \cap B) \cup (A \setminus B) \cup (B \setminus A)$ .

$(\supseteq)$  Let  $x \in (A \cap B) \cup (A \setminus B) \cup (B \setminus A)$  be arbitrary. Then,  $x \in A \cap B$  or  $x \in A \setminus B$  or  $x \in B \setminus A$ . Consider the following cases.

- If  $x \in A \cap B$ , then  $x \in A$  and  $x \in B$  by definition, so  $x \in A \cup B$ .
- If  $x \in A \setminus B$ , then  $x \in A$  by definition, so  $x \in A \cup B$  by definition.
- If  $x \in B \setminus A$ , then  $x \in B$  by definition, so  $x \in A \cup B$  by definition.

Hence, proof by cases,  $(A \cap B) \cup (A \setminus B) \cup (B \setminus A) \subseteq A \cup B$ .

Therefore, by  $(\subseteq)$  and  $(\supseteq)$ ,  $(A \cup B) = (A \cap B) \cup (A \setminus B) \cup (B \setminus A)$ . ■

- (b)  $(A \cap B)$ ,  $(A \setminus B)$ , and  $(B \setminus A)$  are all pairwise disjoint.

**Proof:** It is required to show that any two of these sets have empty intersection,

- To claim  $(A \cap B) \cap (A \setminus B) = \emptyset$ , assume there exists  $x \in (A \cap B) \cap (A \setminus B)$  for the sake of contradiction. Then,  $x \in A \cap B$  and  $x \in A \setminus B$ . Since  $x \in A \cap B$ , it follows that  $x \in A$  and  $x \in B$  by definition of Intersection of Sets. Since  $x \in A \setminus B$ , it follows that  $x \in A$  and  $x \notin B$  by definition of Set Difference, which contradicts  $x \in B$ . Therefore,  $(A \cap B) \cap (A \setminus B) = \emptyset$ .
- To claim  $(A \cap B) \cap (B \setminus A) = \emptyset$ , without loss of generality, it is symmetrical to the previous case, with the roles of sets  $A$  and  $B$  swapped. Hence, trivially,  $(A \cap B) \cap (B \setminus A) = \emptyset$ .
- To claim  $(A \setminus B) \cap (B \setminus A) = \emptyset$ , assume there exists  $x \in (A \setminus B) \cap (B \setminus A)$  for the sake of contradiction. Then,  $x \in A \setminus B$  and  $x \in B \setminus A$ . Since  $x \in A \setminus B$ , it follows that  $x \in A$  and  $x \notin B$  by definition of Set Difference. Since  $x \in B \setminus A$ , it follows that  $x \in B$  and  $x \notin A$  by definition of Set Difference, which contradicts both  $x \notin B$  and  $x \in A$  at the same time. Therefore,  $(A \setminus B) \cap (B \setminus A) = \emptyset$ .

Therefore,  $(A \cap B)$ ,  $(A \setminus B)$ , and  $(B \setminus A)$  are all pairwise disjoint by definition of Pairwise Disjoint Sets. ■

<sup>1</sup>**Notice:** The answers contain cross-references throughout this document as clickable hyperlinks in most PDF viewers. However, the Gradescope PDF view might not support clickable hyperlinks.

<sup>2</sup>I intentionally avoided repetitive or excessive cross-references for better readability.

2. Let  $A$ ,  $B$ ,  $C$ , and  $D$  be subsets of a universal set  $\mathcal{U}$ . Prove or disprove the following propositions.<sup>3</sup>

*Likewise, Cartesian product of sets (above) and the power set of a set (below) will be very important in the next part of the course.*

(a)  $(A \times B) \cap (C \times D) \subseteq (A \cap C) \times (B \cap D)$ .

**Proof:** Let  $(x, y) \in (A \times B) \cap (C \times D)$  be arbitrary for the sake of directly showing  $(\subseteq)$ . Then,  $(x, y) \in A \times B$  and  $(x, y) \in C \times D$ . Since  $(x, y) \in A \times B$ , it follows that  $x \in A$  and  $y \in B$  by definition of Cartesian Product. Since  $(x, y) \in C \times D$ , it follows that  $x \in C$  and  $y \in D$  as well. Thus,  $x \in A \cap C$  and  $y \in B \cap D$  by definition of Intersection of Sets, so  $(x, y) \in (A \cap C) \times (B \cap D)$ . Hence,  $(A \times B) \cap (C \times D) \subseteq (A \cap C) \times (B \cap D)$ . ■

(b)  $(A \cap C) \times (B \cap D) \subseteq (A \times B) \cap (C \times D)$ .

**Proof:** Let  $(x, y) \in (A \cap C) \times (B \cap D)$  be arbitrary for the sake of directly showing  $(\subseteq)$ . Then,  $x \in A \cap C$  and  $y \in B \cap D$  by definition of Cartesian Product. Since  $x \in A \cap C$ , it follows that  $x \in A$  and  $x \in C$  by definition of Intersection of Sets. Since  $y \in B \cap D$ , it follows that  $y \in B$  and  $y \in D$ . Therefore,  $(x, y) \in A \times B$  and  $(x, y) \in C \times D$  since logical AND ( $\wedge$ ) is commutative. Thus,  $(x, y) \in (A \times B) \cap (C \times D)$ . Hence,  $(A \cap C) \times (B \cap D) \subseteq (A \times B) \cap (C \times D)$ . ■

(c)  $(A \times B) \cup (C \times D) \subseteq (A \cup C) \times (B \cup D)$ .

**Proof:** Let  $(x, y) \in (A \times B) \cup (C \times D)$  be arbitrary for the sake of directly showing  $(\subseteq)$ . Then,  $(x, y) \in A \times B$  or  $(x, y) \in C \times D$  by definition of Union of Sets.

- If  $(x, y) \in A \times B$ , then  $x \in A$  and  $y \in B$  by definition of Cartesian Product. Since  $x \in A$ , it follows that  $x \in A \cup C$  by definition of Union of Sets. Since  $y \in B$ , it follows that  $y \in B \cup D$  by definition of Union of Sets. Therefore,  $(x, y) \in (A \cup C) \times (B \cup D)$ .
- If  $(x, y) \in C \times D$ , without loss of generality, it is symmetrical to the previous case. Therefore, it trivially implies  $(x, y) \in (A \cup C) \times (B \cup D)$ .

In both cases,  $(x, y) \in (A \cup C) \times (B \cup D)$ . Hence, proof by cases,  $(A \times B) \cup (C \times D) \subseteq (A \cup C) \times (B \cup D)$ . ■

(d)  $(A \cup C) \times (B \cup D) \subseteq (A \times B) \cup (C \times D)$ .

**Disproof:** Let  $A = \{1\}$ ,  $B = \{2\}$ ,  $C = \{3\}$ ,  $D = \{4\}$  be subsets of a universal set  $\mathcal{U}$  for the sake of counterexample. Then,  $(A \cup C) \times (B \cup D) = \{1, 3\} \times \{2, 4\} = \{(1, 2), (1, 4), (3, 2), (3, 4)\}$  and  $(A \times B) \cup (C \times D) = \{(1, 2)\} \cup \{(3, 4)\} = \{(1, 2), (3, 4)\}$ . Since  $(1, 4) \in (A \cup C) \times (B \cup D)$  but  $(1, 4) \notin (A \times B) \cup (C \times D)$ , it follows that  $(A \cup C) \times (B \cup D) \not\subseteq (A \times B) \cup (C \times D)$ . Therefore, by counterexample, the statement is **false**. ■

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<sup>3</sup>**Notice:** The answers contain cross-references throughout this document as clickable hyperlinks in most PDF viewers. However, the Gradescope PDF view might not support clickable hyperlinks.

3. Let  $A, B$  be subsets of a universal set  $\mathcal{U}$ . Prove or disprove the following propositions.<sup>4</sup>

(a)  $\mathcal{P}(A \cap B) \subseteq \mathcal{P}(A) \cap \mathcal{P}(B)$ .

**Proof:** Let  $X \subseteq \mathcal{U}$ . Assume  $X \in \mathcal{P}(A \cap B)$  be arbitrary for the sake of directly showing  $(\subseteq)$ . Then,  $X \subseteq A \cap B$  by definition of Power Set. Since  $X \subseteq A \cap B$ , it follows that  $X \subseteq A$  and  $X \subseteq B$  by definition of Intersection of Sets. Therefore,  $X \in \mathcal{P}(A)$  and  $X \in \mathcal{P}(B)$ , so  $X \in \mathcal{P}(A) \cap \mathcal{P}(B)$ . Hence,  $\mathcal{P}(A \cap B) \subseteq \mathcal{P}(A) \cap \mathcal{P}(B)$ . ■

(b)  $\mathcal{P}(A) \cap \mathcal{P}(B) \subseteq \mathcal{P}(A \cap B)$ .

**Proof:** Let  $X \subseteq \mathcal{U}$ . Assume  $X \in \mathcal{P}(A) \cap \mathcal{P}(B)$  be arbitrary for the sake of directly showing  $(\subseteq)$ . Then,  $X \in \mathcal{P}(A)$  and  $X \in \mathcal{P}(B)$  by definition of Intersection of Sets. Since  $X \in \mathcal{P}(A)$ , it follows that  $X \subseteq A$  by definition of Power Set. Without loss of generality, (trivial)  $X \subseteq B$ . Since  $X \subseteq A$  and  $X \subseteq B$ , it follows that  $X \subseteq A \cap B$ . Therefore,  $X \in \mathcal{P}(A \cap B)$ . Hence,  $\mathcal{P}(A) \cap \mathcal{P}(B) \subseteq \mathcal{P}(A \cap B)$ . ■

(c)  $\mathcal{P}(A \cup B) \subseteq \mathcal{P}(A) \cup \mathcal{P}(B)$ .

**Disproof:** Let  $A = \{1\}, B = \{2\}$  be subsets of a universal set  $\mathcal{U}$  for the sake of counterexample. Then,  $\mathcal{P}(A \cup B) = \mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$  by definition of Power Set and  $\mathcal{P}(A) \cup \mathcal{P}(B) = \{\emptyset, \{1\}\} \cup \{\emptyset, \{2\}\} = \{\emptyset, \{1\}, \{2\}\}$ . Since  $\{1, 2\} \in \mathcal{P}(A \cup B)$  but  $\{1, 2\} \notin \mathcal{P}(A) \cup \mathcal{P}(B)$ , it follows that  $\mathcal{P}(A \cup B) \not\subseteq \mathcal{P}(A) \cup \mathcal{P}(B)$ . Therefore, by the counterexample, the statement is **false**. ■

(d)  $\mathcal{P}(A) \cup \mathcal{P}(B) \subseteq \mathcal{P}(A \cup B)$ .

**Proof:** Let  $X \subseteq \mathcal{U}$ . Assume  $X \in \mathcal{P}(A) \cup \mathcal{P}(B)$  be arbitrary for the sake of directly showing  $(\subseteq)$ . Then,  $X \in \mathcal{P}(A)$  or  $X \in \mathcal{P}(B)$  by definition of Union of Sets.

§ Subset of Union:  $A \subseteq A \cup B$

**Proof:** Let  $x \in U$ , assume  $x \in A$ . Then the statement “ $x \in A$  or  $x \in B$ ” is true. Hence, by definition of Union of Sets,  $A \cup B = \{x \in U \mid x \in A \text{ or } x \in B\}$ ,  $x \in A \cup B$ . Thus,  $A \subseteq A \cup B$ . ■

If  $X \in \mathcal{P}(A)$ , then  $X \subseteq A \subseteq A \cup B$  by definition of Power Set and the short claim above, Subset of Union:  $A \subseteq A \cup B$ , so  $X \in \mathcal{P}(A \cup B)$ . Without loss of generality, the case of  $X \in \mathcal{P}(B)$  trivially implies  $X \in \mathcal{P}(A \cup B)$  as well. In both cases,  $X \in \mathcal{P}(A \cup B)$ . Hence, proof by cases,  $\mathcal{P}(A) \cup \mathcal{P}(B) \subseteq \mathcal{P}(A \cup B)$ . ■

<sup>4</sup>**Notice:** The answers contain cross-references throughout this document as clickable hyperlinks in most PDF viewers. However, the Gradescope PDF view might not support clickable hyperlinks.

4. [Notice] Each equivalence class should be written exactly once and in set builder notation. For instance, let  $R$  be the relation on  $\mathbb{R}$  given by  $aRb$  if and only if  $a - b \in \mathbb{Z}$ . Then the equivalence classes of  $R$  are  $[a] = \{k + a \mid k \in \mathbb{Z}\}$  for all  $a$  with  $0 \leq a < 1$ . Note that writing  $[a] = \{x \in \mathbb{R} \mid xRa\}$  is unilluminating, and hence not a correct answer.
5. Prove that if  $R$  is an equivalence relation on a nonempty set  $A$ , then the inverse relation  $R^{-1}$  is also an equivalence relation on  $A$ .

**Proof:** Let  $R$  be a relation on a nonempty set  $A$ . To show  $R^{-1}$  is an equivalence relation, we check reflexivity, symmetry, and transitivity.

- **Reflexive:** Since  $R$  is reflexive, for all  $a \in A$ ,  $(a, a) \in R$ . Thus,  $(a, a) \in R^{-1}$ .
- **Symmetric:** If  $(a, b) \in R^{-1}$ , then  $(b, a) \in R$ . Since  $R$  is symmetric,  $(a, b) \in R$ . Thus,  $(b, a) \in R^{-1}$ .
- **Transitive:** If  $(a, b), (b, c) \in R^{-1}$ , then  $(b, a), (c, b) \in R$ . Since  $R$  is transitive,  $(c, a) \in R$ , so  $(a, c) \in R^{-1}$ .

Therefore,  $R^{-1}$  is an equivalence relation. ■

6. Suppose that  $R$  and  $S$  are equivalence relations on the nonempty set  $A$ . Prove or disprove:

- (a) The set  $R \cup S$  is an equivalence relation.

**Disproof:** Let  $A = \{1, 2, 3\}$ ,  $R, S$  be a relation on  $A$  where  $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$  and  $S = \{(2, 2), (3, 3), (2, 3), (3, 2)\}$ . Then,  $(1, 2) \in R \cup S$ ,  $(2, 3) \in R \cup S$ , but  $(1, 3) \notin R \cup S$ . Thus,  $R \cup S$  is not always an equivalence relation. ■

- (b) The set  $R \cap S$  is an equivalence relation.

**Proof:**  $R \cap S$  is reflexive, symmetric, and transitive since both  $R$  and  $S$  are equivalence relations and intersection preserves these properties. ■

7. Let  $R$  be a relation from a set  $A$  to a set  $B$ .

- (a) Prove that a relation  $R$  on a set  $A$  is symmetric if and only if  $R = R^{-1}$ .

**Proof:**  $(\Rightarrow)$  If  $R$  is symmetric, then for all  $(a, b) \in R$ ,  $(b, a) \in R$ , so  $R = R^{-1}$ .  $(\Leftarrow)$  If  $R = R^{-1}$ , then for all  $(a, b) \in R$ ,  $(b, a) \in R$ , so  $R$  is symmetric. ■

- (b) Prove that a relation  $R$  on a set  $A$  is antisymmetric if and only if  $R \cap R^{-1}$  is a subset of the diagonal relation  $\Delta = \{(a, a) \mid a \in A\}$ .

**Proof:**  $(\Rightarrow)$  If  $R$  is antisymmetric, then  $(a, b) \in R$  and  $(b, a) \in R$  implies  $a = b$ , so  $(a, b) \in \Delta$ .  $(\Leftarrow)$  If  $R \cap R^{-1} \subseteq \Delta$ , then  $(a, b) \in R$  and  $(b, a) \in R$  implies  $a = b$ , so  $R$  is antisymmetric. ■

8. Let  $R$  be a relation on the nonempty set  $A$ . Then  $R$  is *irreflexive* iff for every  $a \in A$ ,  $(a, a) \notin R$ . Prove or disprove:

(a) The relation  $R$  could be neither reflexive nor irreflexive.

**Proof:** True. For example,  $A = \{1, 2\}$ ,  $R = \{(1, 1)\}$ .  $(1, 1) \in R$  but  $(2, 2) \notin R$ , so  $R$  is neither reflexive nor irreflexive. ■

(b) The relation  $R$  is reflexive if and only if  $\overline{R}$  is irreflexive.

**Proof:** If  $R$  is reflexive, then for all  $a \in A$ ,  $(a, a) \in R$ , so  $(a, a) \notin \overline{R}$ , so  $\overline{R}$  is irreflexive. Conversely, if  $\overline{R}$  is irreflexive, then for all  $a \in A$ ,  $(a, a) \notin \overline{R}$ , so  $(a, a) \in R$ , so  $R$  is reflexive. ■

(c) If  $R$  is irreflexive, then so is  $R^2 = R \circ R$ .

**Disproof:** False. Let  $A = \{1, 2\}$ ,  $R = \{(1, 2), (2, 1)\}$ , which is irreflexive. But  $R^2 = \{(1, 1), (2, 2)\}$ , which is not irreflexive. ■

9. Let  $R$ ,  $S$ , and  $T$  be relations on  $\mathbb{Z}$  defined as:

- $aRb$  iff  $a \equiv b \pmod{2}$ ,
- $aSb$  iff  $a \equiv b \pmod{3}$ ,
- $aTb$  iff  $a \equiv b \pmod{6}$ .

(a) List the equivalence classes for  $R$ ,  $S$ , and  $T$ .

**Answer:**  $R$ :  $\{[0]_2, [1]_2\}$ , where  $[0]_2 = \{\dots, -2, 0, 2, \dots\}$ ,  $[1]_2 = \{\dots, -1, 1, 3, \dots\}$   
 $S$ :  $\{[0]_3, [1]_3, [2]_3\}$   
 $T$ :  $\{[0]_6, [1]_6, [2]_6, [3]_6, [4]_6, [5]_6\}$

□

(b) Find the equivalence classes of  $R \cap S$ .

**Answer:**  $R \cap S$  is congruence mod 6, so the equivalence classes are  $[0]_6, [1]_6, [2]_6, [3]_6, [4]_6, [5]_6$ .

□

(c) Prove the partition resulting from  $T$  is a refinement of the partition resulting from  $S$ . *Definition: A partition  $P_1$  is a refinement of the partition  $P_2$  if every set in  $P_1$  is a subset of one of the sets in  $P_2$ .*

**Proof:** Each  $[k]_6$  is contained in some  $[j]_3$  since  $a \equiv k \pmod{6}$  implies  $a \equiv k \pmod{3}$ . Thus, every  $[k]_6 \subseteq [j]_3$  for some  $j$ .

■

(d) What is the relationship between the partition resulting from  $T$  and the partition resulting from  $R$ ? Justify your answer.

**Answer:** Each  $[k]_6$  is contained in either  $[0]_2$  or  $[1]_2$  since  $a \equiv k \pmod{6}$  implies  $a \equiv k \pmod{2}$ . So  $T$ 's partition refines  $R$ 's partition.

□

(e) What is the relationship between  $S$  and  $T$ ? Between  $R$  and  $T$ ? Justify your answers.

*Hint: remember that a relation is just a set, and that common relations between sets are "is a subset of", "is a proper subset of", and "is equal to."*

**Answer:**  $T \subseteq S$  and  $T \subseteq R$  since congruence mod 6 implies congruence mod 3 and mod 2.

□

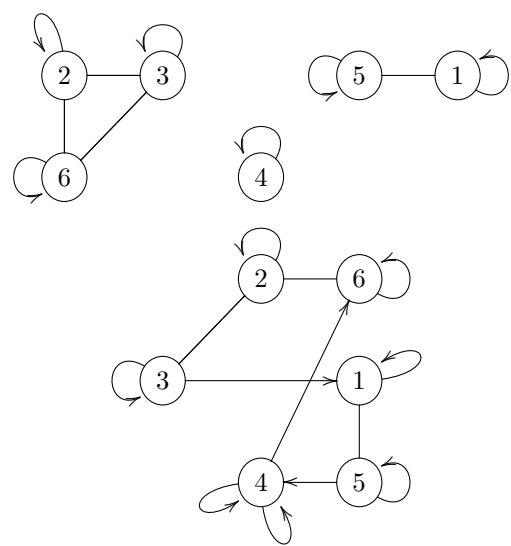
(f) Is there any relationship between  $R$  and  $S$ ? Justify your answers.

**Answer:**  $R$  and  $S$  are not subsets of each other and are not equal; they are independent equivalence relations.

□



**Equivalence Classes and Relation Graph**



**Equivalence Classes and Relation Graph**

## Lecture 5: Dirichlet's Theorem and applications of the FTA

**§ Theorem 1.21 (Dirichlet's theorem on primes in arithmetic progressions)** Let  $a, b \in \mathbb{Z}$  with  $b > 0$  and  $\gcd(a, b) = 1$ . Then the arithmetic progression

$$a, a + b, a + 2b, a + 3b, \dots$$

contains infinitely many prime numbers.

**§ Remark (on Theorem Theorem 1.21 (Dirichlet's theorem on primes in arithmetic progressions))** Setting  $a = 1$  and  $b = 1$  recovers the elementary statement that there are infinitely many primes. (We remark that the full proof of Dirichlet's theorem requires analytic tools and is beyond the scope of this course.)

**§ Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4)** There are infinitely many primes of the form  $4n + 3$ , where  $n$  is a nonnegative integer.

**§ Lemma 1.23 (Product of numbers congruent to 1 modulo 4)** Let  $a, b \in \mathbb{Z}$ . If  $a \equiv 1 \pmod{4}$  and  $b \equiv 1 \pmod{4}$ , then  $ab \equiv 1 \pmod{4}$ .

*Proof.* Write  $a = 4n_1 + 1$  and  $b = 4n_2 + 1$  with  $n_1, n_2 \in \mathbb{Z}$ . Then

$$ab = (4n_1 + 1)(4n_2 + 1) = 16n_1n_2 + 4n_1 + 4n_2 + 1 = 4(4n_1n_2 + n_1 + n_2) + 1,$$

so  $ab \equiv 1 \pmod{4}$ . This proves the lemma.  $\square$

*Proof of Proposition Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4).* Suppose, for contradiction, that there are only finitely many primes congruent to 3 modulo 4. List them as

$$P_1 = 3, P_2, P_3, \dots, P_r,$$

all distinct and each  $P_i \equiv 3 \pmod{4}$ .

Consider the integer

$$N := 4P_1P_2 \cdots P_r - 1.$$

Note that  $N \equiv -1 \equiv 3 \pmod{4}$ .

Let  $q$  be any prime divisor of  $N$ . We make two observations:

1. For each  $i = 1, \dots, r$  we have  $N \equiv -1 \pmod{P_i}$ , so  $P_i \nmid N$ . Thus  $q$  is not equal to any of the  $P_i$ .
2. Since  $N \equiv 3 \pmod{4}$ , it is not possible that every prime divisor of  $N$  is congruent to 1 modulo 4. Indeed, a product of integers each congruent to 1 modulo 4 is itself congruent to 1 modulo 4 by Lemma Lemma 1.23 (Product of numbers congruent to 1 modulo 4), contradicting  $N \equiv 3 \pmod{4}$ . Hence some prime divisor  $q$  of  $N$  satisfies  $q \equiv 3 \pmod{4}$ .

Combining the two observations,  $q$  is a prime congruent to 3 modulo 4 that is not among  $P_1, \dots, P_r$ , contradicting the assumed finiteness of such primes. Therefore there are infinitely many primes congruent to 3 modulo 4.  $\square$

**§ Remark (on Proposition Proposition 1.22 (Infinitely many primes congruent to 3 modulo 4))** The method above—building an integer with residue 3 modulo 4 whose prime divisors cannot all be from a given finite list—gives a proof in this special case of Dirichlet's theorem. However, this argument does not extend directly to all arithmetic progressions; for the full theorem one needs deeper tools (Dirichlet characters and  $L$ -series).

## Congruences

**§ Historical remark (Gauss and congruences)** Up to the beginning of the nineteenth century number theory consisted of many isolated results. In 1801, Gauss, in his *Disquisitiones Arithmeticae*, introduced the modern theory of congruences and unified many isolated results in number theory.

**§ Definition (Congruence modulo  $m$ )** Let  $a, b, m \in \mathbb{Z}$  with  $m > 0$ . We say that  $a$  is congruent to  $b$  modulo  $m$ , and write

$$a \equiv b \pmod{m},$$

if  $m \mid (a - b)$ . The integer  $m$  is called the *modulus* of the congruence.

**§ Remark (notation)** The notation  $a \not\equiv b \pmod{m}$  means  $m \nmid (a - b)$ . The modulus is customarily taken to be positive; we will assume  $m > 0$  in the sequel.

**§ Proposition 2.1 (Congruence modulo  $m$  is an equivalence relation)** Congruence modulo  $m$  is reflexive, symmetric, and transitive on  $\mathbb{Z}$ ; hence it is an equivalence relation.

*Proof. Reflexive:* For any  $a \in \mathbb{Z}$ ,  $a - a = 0$  and  $m \mid 0$ , so  $a \equiv a \pmod{m}$ .

*Symmetric:* If  $a \equiv b \pmod{m}$  then  $m \mid (a - b)$ , hence  $m \mid -(a - b) = (b - a)$ , so  $b \equiv a \pmod{m}$ .

*Transitive:* If  $a \equiv b \pmod{m}$  and  $b \equiv c \pmod{m}$  then  $m \mid (a - b)$  and  $m \mid (b - c)$ , therefore  $m \mid (a - b) + (b - c) = a - c$ , so  $a \equiv c \pmod{m}$ .  $\square$

**§ Consequence 2.2 (Partition into residue classes)** The set  $\mathbb{Z}$  is partitioned into equivalence classes under congruence modulo  $m$ .

**§ Definition (Notation for residue class)** For  $x \in \mathbb{Z}$ , write  $[x]_m$  (or simply  $[x]$  when  $m$  is understood) for the equivalence class of  $x$  modulo  $m$ :

$$[x]_m = \{y \in \mathbb{Z} : y \equiv x \pmod{m}\}.$$

(Do not confuse  $[x]_m$  with the floor function.)

### Example: The equivalence classes modulo 4

$$[0]_4 = \{x \in \mathbb{Z} : x \equiv 0 \pmod{4}\} = \{4n : n \in \mathbb{Z}\} = \{\dots, -8, -4, 0, 4, 8, \dots\},$$

$$[1]_4 = \{x : x = 4n + 1, n \in \mathbb{Z}\} = \{\dots, -7, -3, 1, 5, 9, \dots\},$$

$$[2]_4 = \{x : x = 4n + 2, n \in \mathbb{Z}\} = \{\dots, -6, -2, 2, 6, 10, \dots\},$$

$$[3]_4 = \{x : x = 4n + 3, n \in \mathbb{Z}\} = \{\dots, -5, -1, 3, 7, 11, \dots\}.$$

Thus  $\mathbb{Z}$  is partitioned into four residue classes modulo 4, and every integer is congruent to exactly one of 0, 1, 2, 3 modulo 4.

**§ Definition (Complete residue system modulo  $m$ )** A set  $R \subset \mathbb{Z}$  is a *complete residue system modulo  $m$*  if every integer is congruent modulo  $m$  to exactly one element of  $R$ .

**§ Example (complete residue systems)** The set  $\{0, 1, 2, \dots, m - 1\}$  is a complete residue system modulo  $m$ . Also, for example,  $\{6, -11, 19, 1988\}$  is a complete residue system modulo 4, since  $6 \equiv 2$ ,  $-11 \equiv 1$ ,  $19 \equiv 3$ , and  $1988 \equiv 0 \pmod{4}$ .

**§ Proposition 2.3 (Standard complete residue system)** The set  $\{0, 1, 2, \dots, m - 1\}$  is a complete residue system modulo  $m$ .

*Proof.* Let  $a \in \mathbb{Z}$ . By the Division Algorithm there exist unique  $q, r \in \mathbb{Z}$  with  $0 \leq r < m$  such that  $a = mq + r$ . Then  $a - r = mq$ , so  $m \mid (a - r)$  and  $a \equiv r \pmod{m}$ , with  $r \in \{0, 1, \dots, m - 1\}$ . This shows every integer is congruent to at least one element of  $\{0, 1, \dots, m - 1\}$ .

To show uniqueness: suppose  $r_1, r_2 \in \{0, 1, \dots, m-1\}$  and  $r_1 \equiv r_2 \pmod{m}$ . Then  $m \mid (r_1 - r_2)$ , but  $-(m-1) \leq r_1 - r_2 \leq m-1$ . The only multiple of  $m$  in this range is 0, so  $r_1 - r_2 = 0$  and hence  $r_1 = r_2$ . Thus each integer is congruent to exactly one element of  $\{0, 1, \dots, m-1\}$ .  $\square$

**§ Proposition 2.4 (Arithmetic with congruences)** Let  $a, b, c, d \in \mathbb{Z}$  and  $m > 0$ . If  $a \equiv b \pmod{m}$  and  $c \equiv d \pmod{m}$ , then

1.  $a + c \equiv b + d \pmod{m}$ ,
2.  $ac \equiv bd \pmod{m}$ .

*Proof.* From  $a \equiv b \pmod{m}$  and  $c \equiv d \pmod{m}$  we have  $m \mid (a - b)$  and  $m \mid (c - d)$ .

For (1):  $m \mid (a - b) + (c - d) = (a + c) - (b + d)$ , hence  $a + c \equiv b + d \pmod{m}$ .

For (2): Note  $m \mid (a - b)$  implies  $m \mid c(a - b)$ , and  $m \mid (c - d)$  implies  $m \mid b(c - d)$ . Add these divisibilities to get  $m \mid c(a - b) + b(c - d) = ac - bd$ , so  $ac \equiv bd \pmod{m}$ .  $\square$

## L6: Congruences

**§ Remark (Proposition 2.4 (Arithmetic with congruences) in class terms)** Recall that Proposition 2.4 (Arithmetic with congruences) from the previous lecture can be expressed in terms of equivalence classes modulo  $m$ : for residue classes  $[a], [c] \in \mathbb{Z}_m$ ,

$$[a] + [c] = [a + c] \quad \text{and} \quad [a][c] = [ac].$$

This viewpoint allows construction of addition and multiplication tables for residue classes (modulo  $m$ ) using any complete residue system.

### Examples: tables for $\mathbb{Z}_4$ and $\mathbb{Z}_5$

| +       | $[0]_4$ | $[1]_4$ | $[2]_4$ | $[3]_4$ |
|---------|---------|---------|---------|---------|
| $[0]_4$ | $[0]_4$ | $[1]_4$ | $[2]_4$ | $[3]_4$ |
| $[1]_4$ | $[1]_4$ | $[2]_4$ | $[3]_4$ | $[0]_4$ |
| $[2]_4$ | $[2]_4$ | $[3]_4$ | $[0]_4$ | $[1]_4$ |
| $[3]_4$ | $[3]_4$ | $[0]_4$ | $[1]_4$ | $[2]_4$ |

| ·       | $[0]_4$ | $[1]_4$ | $[2]_4$ | $[3]_4$ |
|---------|---------|---------|---------|---------|
| $[0]_4$ | $[0]_4$ | $[0]_4$ | $[0]_4$ | $[0]_4$ |
| $[1]_4$ | $[0]_4$ | $[1]_4$ | $[2]_4$ | $[3]_4$ |
| $[2]_4$ | $[0]_4$ | $[2]_4$ | $[0]_4$ | $[2]_4$ |
| $[3]_4$ | $[0]_4$ | $[3]_4$ | $[2]_4$ | $[1]_4$ |

**§ Remark (structure of  $\mathbb{Z}_4$ )** Set  $\mathbb{Z}_4 := \{[0]_4, [1]_4, [2]_4, [3]_4\}$ . With the above addition and multiplication it is a ring, but not a field: not every nonzero element has a multiplicative inverse (e.g.,  $[2]_4$  is a zero divisor).

| +       | $[0]_5$ | $[1]_5$ | $[2]_5$ | $[3]_5$ | $[4]_5$ |
|---------|---------|---------|---------|---------|---------|
| $[0]_5$ | $[0]_5$ | $[1]_5$ | $[2]_5$ | $[3]_5$ | $[4]_5$ |
| $[1]_5$ | $[1]_5$ | $[2]_5$ | $[3]_5$ | $[4]_5$ | $[0]_5$ |
| $[2]_5$ | $[2]_5$ | $[3]_5$ | $[4]_5$ | $[0]_5$ | $[1]_5$ |
| $[3]_5$ | $[3]_5$ | $[4]_5$ | $[0]_5$ | $[1]_5$ | $[2]_5$ |
| $[4]_5$ | $[4]_5$ | $[0]_5$ | $[1]_5$ | $[2]_5$ | $[3]_5$ |

| ·       | $[0]_5$ | $[1]_5$ | $[2]_5$ | $[3]_5$ | $[4]_5$ |
|---------|---------|---------|---------|---------|---------|
| $[0]_5$ | $[0]_5$ | $[0]_5$ | $[0]_5$ | $[0]_5$ | $[0]_5$ |
| $[1]_5$ | $[0]_5$ | $[1]_5$ | $[2]_5$ | $[3]_5$ | $[4]_5$ |
| $[2]_5$ | $[0]_5$ | $[2]_5$ | $[4]_5$ | $[1]_5$ | $[3]_5$ |
| $[3]_5$ | $[0]_5$ | $[3]_5$ | $[1]_5$ | $[4]_5$ | $[2]_5$ |
| $[4]_5$ | $[0]_5$ | $[4]_5$ | $[3]_5$ | $[2]_5$ | $[1]_5$ |

**§ Remark (structure of  $\mathbb{Z}_5$ )** Let  $\mathbb{Z}_5 := \{[0]_5, [1]_5, [2]_5, [3]_5, [4]_5\}$ . Because 5 is prime every nonzero class has a multiplicative inverse, so  $\mathbb{Z}_5$  is a field.

**§ Question (why extra structure?)** Why does  $\mathbb{Z}_5$  have more structure (a field) than  $\mathbb{Z}_4$  (only a ring)?

**§ Answer (primality of modulus)** Essentially:  $\mathbb{Z}_m$  is a field if and only if  $m$  is prime. If  $m$  is composite, then some nonzero classes are zero divisors (no inverses). If  $m = p$  is prime, then every nonzero class has an inverse (standard proof via Bezout's identity).

### Cancellation and a useful proposition

**§ Proposition 2.5 (Cancellation in congruences)** Let  $a, b, c, m \in \mathbb{Z}$  with  $m > 0$ , and let  $d = \gcd(c, m)$ . Then

$$ca \equiv cb \pmod{m} \iff a \equiv b \pmod{m/d}.$$

*Proof.* Suppose  $ca \equiv cb \pmod{m}$ , i.e.  $m \mid c(a - b)$ . Let  $d = \gcd(c, m)$ . Write  $c = d \cdot c'$  and  $m = d \cdot m'$  so that  $\gcd(c', m') = 1$ . From  $m \mid c(a - b)$  we get  $dm' \mid dc'(a - b)$ , hence  $m' \mid c'(a - b)$ . Since  $\gcd(c', m') = 1$ , it follows that  $m' \mid (a - b)$ , i.e.  $a \equiv b \pmod{m'}$ , which is  $a \equiv b \pmod{m/d}$ .

Conversely, if  $a \equiv b \pmod{m/d}$  then  $m/d \mid (a - b)$ , so  $m/d \cdot d = m \mid d(a - b)$ . But  $d \mid c$ , so  $c(a - b)$  is a multiple of  $d(a - b)$ , hence  $m \mid c(a - b)$  and  $ca \equiv cb \pmod{m}$ . This proves the equivalence.  $\square$

## Linear congruences in one variable

**§ Definition (linear congruence)** A congruence of the form

$$ax \equiv b \pmod{m},$$

with unknown  $x \in \mathbb{Z}$  and given integers  $a, b, m$  with  $m > 0$ , is called a *linear congruence* in the variable  $x$ .

**§ Examples (small checks)**

- $2x \equiv 3 \pmod{4}$  has no solution: checking  $x \equiv 0, 1, 2, 3 \pmod{4}$  yields no solution.
- $2x \equiv 3 \pmod{5}$  has the unique solution  $x \equiv 4 \pmod{5}$ .
- $2x \equiv 4 \pmod{6}$  has two incongruent solutions modulo 6:  $x \equiv 2, 5 \pmod{6}$ .
- $3x \equiv 9 \pmod{6}$  has three incongruent solutions modulo 6:  $x \equiv 1, 3, 5 \pmod{6}$ .

**§ Theorem 2.6 (Solvability and number of solutions of linear congruences)** Let  $a, b, m \in \mathbb{Z}$  with  $m > 0$  and let  $d = \gcd(a, m)$ . Then:

1. The congruence  $ax \equiv b \pmod{m}$  has a solution iff  $d \mid b$ .
2. If  $d \mid b$  then there are exactly  $d$  incongruent solutions modulo  $m$ . Concretely, if  $x_0$  is one solution, then all solutions are

$$x = x_0 + \frac{m}{d}n, \quad n \in \mathbb{Z},$$

and the classes  $x_0 + \frac{m}{d}n$  for  $n = 0, 1, \dots, d-1$  give a complete set of incongruent solutions modulo  $m$ .

*Proof.* We use the standard equivalence between the congruence and a linear Diophantine equation. The congruence  $ax \equiv b \pmod{m}$  is equivalent to the existence of an integer  $y$  such that

$$ax - my = b. \quad (*)$$

Thus solvability of the congruence is equivalent to solvability of the linear Diophantine equation (\*).

(1) If (\*) has an integer solution, then  $d = \gcd(a, m)$  divides the left-hand side  $ax - my$  and hence divides  $b$ . Conversely, if  $d \mid b$ , write  $b = de$  for some  $e \in \mathbb{Z}$ . By Bezout's identity (see ?? or the standard result), there exist integers  $r, s$  with  $ar + ms = d$ . Multiply both sides by  $e$  to obtain  $are + mse = de = b$ . Then  $x_0 = re$  and  $y_0 = -se$  solve (\*), so a solution exists.

(2) Fix any solution  $x_0$  of  $ax \equiv b \pmod{m}$ . For any integer  $n$  consider

$$x = x_0 + \frac{m}{d}n.$$

Then

$$ax = ax_0 + a \frac{m}{d}n = ax_0 + \frac{a}{d}mn \equiv ax_0 \equiv b \pmod{m},$$

so each such  $x$  is a solution. Thus there are infinitely many integer solutions, all congruent modulo  $\frac{m}{d}$  shifts.

Now suppose  $x$  is any solution. Then subtracting the equation for  $x_0$  from the equation for  $x$  yields

$$a(x - x_0) \equiv 0 \pmod{m},$$

i.e.  $m \mid a(x - x_0)$ . Write  $a = da'$  and  $m = dm'$ , with  $\gcd(a', m') = 1$ . Then  $m'd \mid da'(x - x_0)$ , so  $m' \mid a'(x - x_0)$ . Since  $\gcd(a', m') = 1$  we deduce  $m' \mid (x - x_0)$ , hence  $x - x_0 = m'n = \frac{m}{d}n$  for some  $n \in \mathbb{Z}$ . This shows every solution has the stated form. Two such solutions

$$x_0 + \frac{m}{d}n_1 \quad \text{and} \quad x_0 + \frac{m}{d}n_2$$

are congruent modulo  $m$  iff

$$\frac{m}{d}(n_1 - n_2) \equiv 0 \pmod{m} \iff m \mid \frac{m}{d}(n_1 - n_2).$$

Dividing both sides by  $m/d$  yields  $d \mid (n_1 - n_2)$ , so  $n_1 \equiv n_2 \pmod{d}$ . Therefore choosing  $n = 0, 1, \dots, d-1$  produces exactly  $d$  incongruent solutions modulo  $m$ .  $\square$

**§ Corollary 2.7 (Explicit solution set)** Under the hypotheses of Theorem Theorem 2.6 (Solvability and number of solutions of linear congruences), if  $d = \gcd(a, m)$  and  $d \mid b$ , then the  $d$  incongruent solutions modulo  $m$  are

$$x_0 + \frac{m}{d}n, \quad n = 0, 1, \dots, d-1,$$

where  $x_0$  is any particular solution.

## Worked example

**§ Example (solve  $16x \equiv 8 \pmod{28}$ )** We solve  $16x \equiv 8 \pmod{28}$ . Compute  $d = \gcd(16, 28)$ .

Euclidean algorithm:

$$28 = 1 \cdot 16 + 12, \quad 16 = 1 \cdot 12 + 4, \quad 12 = 3 \cdot 4 + 0,$$

so  $d = 4$ . Since  $4 \mid 8$ , the congruence has  $d = 4$  incongruent solutions modulo 28 by Theorem Theorem 2.6 (Solvability and number of solutions of linear congruences).

One method: divide the congruence by  $d = 4$  to get (working modulo  $28/4 = 7$ )

$$4x \equiv 2 \pmod{7}.$$

Since  $4^{-1} \equiv 2 \pmod{7}$  (because  $4 \cdot 2 = 8 \equiv 1 \pmod{7}$ ), multiply both sides by 2:

$$x \equiv 4 \pmod{7}.$$

Thus the solutions modulo 28 are

$$x \equiv 4 + 7n, \quad n = 0, 1, 2, 3,$$

which gives the incongruent solutions  $x \equiv 4, 11, 18, 25 \pmod{28}$ .

(Alternatively, one can find an explicit  $x_0$  via the extended Euclidean computation: from  $4 = 2 \cdot 16 - 1 \cdot 28$  we get  $8 = 4 \cdot 16 - 2 \cdot 28$ , so  $x_0 = 4$  is a particular solution, yielding the same solution set.)

## Multiplicative inverses

**§ Definition (multiplicative inverse modulo  $m$ )** If  $a \in \mathbb{Z}$  and there exists  $x \in \mathbb{Z}$  with  $ax \equiv 1 \pmod{m}$ , then  $x$  is called a *multiplicative inverse* of  $a$  modulo  $m$  (often denoted  $a^{-1} \pmod{m}$ ).

**§ Corollary 2.8 (When inverses exist)** The congruence  $ax \equiv 1 \pmod{m}$  has a solution iff  $\gcd(a, m) = 1$ . In that case the inverse is unique modulo  $m$ .

*Proof.* Apply Theorem Theorem 2.6 (Solvability and number of solutions of linear congruences) with  $b = 1$ . If  $d = \gcd(a, m) > 1$  then  $d \nmid 1$  so there is no solution. If  $d = 1$  then there is exactly one solution modulo  $m$ .  $\square$

**§ Example (solve  $5x \equiv 12 \pmod{16}$ )** Since  $\gcd(5, 16) = 1$ , the congruence has a unique solution modulo 16 by Corollary Corollary 2.8 (When inverses exist). One checks that  $5^{-1} \pmod{16} = 13$  because  $5 \cdot 13 = 65 \equiv 1 \pmod{16}$ . Multiplying by 13 gives

$$x \equiv 13 \cdot 12 \pmod{16}.$$

Compute  $13 \cdot 12 = 156 \equiv 12 \pmod{16}$  (since  $156 - 144 = 12$ ), so  $x \equiv 12 \pmod{16}$ .

# L7: The Chinese Remainder Theorem, Wilson's Theorem, Fermat's Little Theorem

**Example (motivation).** Find a positive integer  $x$  with

$$x \equiv 2 \pmod{3}, \quad x \equiv 1 \pmod{4}, \quad x \equiv 3 \pmod{5}.$$

This is a system of linear congruences in one variable.

**§ Theorem 2.9 (Chinese Remainder Theorem)** Let  $m_1, m_2, \dots, m_n$  be pairwise relatively prime positive integers, and let  $b_1, \dots, b_n$  be arbitrary integers. Then the system

$$x \equiv b_i \pmod{m_i} \quad (i = 1, \dots, n)$$

has a solution, and any two solutions are congruent modulo  $M := m_1 m_2 \cdots m_n$ . Hence the solution is unique modulo  $M$ .

*Proof.* Let  $M := \prod_{i=1}^n m_i$  and define  $M_i := M/m_i$ . Since  $(M_i, m_i) = 1$  by the pairwise coprimality hypothesis, by Corollary 2.8 (When inverses exist) (existence of inverses when gcd is 1) there exists  $x_i$  with

$$M_i x_i \equiv 1 \pmod{m_i}.$$

Put

$$x := \sum_{i=1}^n b_i M_i x_i.$$

For a fixed index  $i$ , reduce modulo  $m_i$ : every term  $b_j M_j x_j$  with  $j \neq i$  is a multiple of  $m_i$  (since  $m_i \mid M_j$  for  $j \neq i$ ), while  $M_i x_i \equiv 1 \pmod{m_i}$  by construction. Thus  $x \equiv b_i \pmod{m_i}$  for each  $i$ , so  $x$  is a solution.

If  $x'$  is another solution, then  $m_i \mid (x - x')$  for every  $i$ , and since the  $m_i$  are pairwise coprime it follows that  $M \mid (x - x')$ . Hence  $x \equiv x' \pmod{M}$ , proving uniqueness modulo  $M$ .  $\square$

**Example (continued).** Here  $m_1 = 3, m_2 = 4, m_3 = 5$ , so  $M = 60$  and

$$M_1 = 60/3 = 20, \quad M_2 = 60/4 = 15, \quad M_3 = 60/5 = 12.$$

Solve  $M_i x_i \equiv 1 \pmod{m_i}$ :

$$20x_1 \equiv 1 \pmod{3} \implies 2x_1 \equiv 1 \pmod{3} \implies x_1 \equiv 2 \pmod{3},$$

so choose  $x_1 = 2$ .

$$15x_2 \equiv 1 \pmod{4} \implies 3x_2 \equiv 1 \pmod{4} \implies x_2 \equiv 3 \pmod{4},$$

so choose  $x_2 = 3$ .

$$12x_3 \equiv 1 \pmod{5} \implies 2x_3 \equiv 1 \pmod{5} \implies x_3 \equiv 3 \pmod{5},$$

so choose  $x_3 = 3$ .

Now

$$x = b_1 M_1 x_1 + b_2 M_2 x_2 + b_3 M_3 x_3 = 2 \cdot 20 \cdot 2 + 1 \cdot 15 \cdot 3 + 3 \cdot 12 \cdot 3 = 80 + 45 + 108 = 233.$$

Hence  $x \equiv 233 \equiv 53 \pmod{60}$ , and the least positive solution is 53.

**§ Lemma 2.10 (Self-inverse modulo a prime)** Let  $p$  be a prime and  $a \in \mathbb{Z}$ . Then  $a$  is its own inverse modulo  $p$  (i.e.  $a^2 \equiv 1 \pmod{p}$ ) if and only if  $a \equiv \pm 1 \pmod{p}$ .



*Proof.* If  $a^2 \equiv 1 \pmod{p}$  then  $p \mid (a-1)(a+1)$ . Since  $p$  is prime,  $p \mid (a-1)$  or  $p \mid (a+1)$ , so  $a \equiv 1$  or  $a \equiv -1 \pmod{p}$ . The converse is immediate.  $\square$

**§ Theorem 2.11 (Wilson's Theorem)** Let  $p$  be a prime. Then

$$(p-1)! \equiv -1 \pmod{p}.$$

*Proof.* The result is trivial for  $p = 2, 3$ , so assume  $p \geq 5$ . The integers  $1, 2, \dots, p-1$  can be paired into multiplicative inverses modulo  $p$ . By Lemma , the only elements that are their own inverses are 1 and  $p-1$ . Thus, aside from 1 and  $p-1$ , the remaining  $p-3$  elements pair off, and each pair multiplies to 1 modulo  $p$ . Therefore

$$(p-1)! \equiv 1 \cdot (\text{product of inverse-pairs}) \cdot (p-1) \equiv 1 \cdot 1 \cdots 1 \cdot (-1) \equiv -1 \pmod{p},$$

as required.  $\square$

**§ Remark (converse)** The converse is also true: if  $n > 1$  and  $(n-1)! \equiv -1 \pmod{n}$  then  $n$  must be prime.

**§ Proposition 2.12 (Converse to Wilson's Theorem)** Let  $n > 1$  be an integer. If  $(n-1)! \equiv -1 \pmod{n}$  then  $n$  is prime.

*Proof.* Suppose  $n$  is composite, so  $n = ab$  with  $1 < a \leq b < n$ . Then  $a \mid (n-1)!$  because  $(n-1)!$  contains the factor  $a$ , and hence  $(n-1)! \equiv 0 \pmod{a}$ . The hypothesis  $(n-1)! \equiv -1 \pmod{n}$  implies  $(n-1)! \equiv -1 \pmod{a}$  (since  $a \mid n$ ), so  $-1 \equiv 0 \pmod{a}$ , which is impossible. Therefore no such nontrivial factor  $a$  exists and  $n$  must be prime.  $\square$

**§ Remark (Wilson primes)** A prime  $p$  is called a *Wilson prime* if  $(p-1)! \equiv -1 \pmod{p^2}$ . Only three Wilson primes are known: 5, 13, 563. It is unknown whether infinitely many (or only finitely many) Wilson primes exist.

**§ Theorem 2.13 (Fermat's Little Theorem)** Let  $p$  be a prime and let  $a \in \mathbb{Z}$  with  $p \nmid a$ . Then

$$a^{p-1} \equiv 1 \pmod{p}.$$

*Proof.* Consider the  $p-1$  integers  $a, 2a, \dots, (p-1)a$ . None of these is congruent to 0  $\pmod{p}$  (since  $p \nmid a$ ), and no two are congruent modulo  $p$  (if  $ia \equiv ja \pmod{p}$  with  $1 \leq i < j \leq p-1$ , multiply by the inverse of  $a$  modulo  $p$ , which exists by Corollary 2.8 (When inverses exist), to deduce  $i \equiv j \pmod{p}$ , impossible). Hence the residues of  $a, 2a, \dots, (p-1)a$  modulo  $p$  are a permutation of  $1, 2, \dots, p-1$ . Multiplying all these congruences yields

$$a^{p-1}(p-1)! \equiv (p-1)! \pmod{p}.$$

Since  $p \nmid (p-1)!$  (true for prime  $p$ ), we may cancel  $(p-1)!$  from both sides to obtain  $a^{p-1} \equiv 1 \pmod{p}$ . (Note: this cancellation step uses only that  $\gcd((p-1)!, p) = 1$ ; Wilson's Theorem is not required.)  $\square$

## L8: Corollaries to Fermat's Little Theorem, Pseudoprimes, Euler's Theorem, Arithmetic Functions

**§ Corollary 2.14 (Inverse modulo a prime)** Let  $p$  be a prime and let  $a \in \mathbb{Z}$  with  $p \nmid a$ . Then  $a^{p-2}$  is the multiplicative inverse of  $a$  modulo  $p$ ; that is,

$$a \cdot a^{p-2} \equiv 1 \pmod{p}.$$

*Proof.* By Fermat's Little Theorem (Theorem ),  $a^{p-1} \equiv 1 \pmod{p}$ . Multiply both sides by  $a^{-1}$  if you like, or simply rewrite  $a^{p-1} = a \cdot a^{p-2}$  to get  $a \cdot a^{p-2} \equiv 1 \pmod{p}$ . Thus  $a^{p-2}$  is an inverse of  $a$  modulo  $p$ .  $\square$

**§ Remark (application of Corollary Corollary 2.14 (Inverse modulo a prime))** Corollary Corollary 2.14 (Inverse modulo a prime) is useful for solving linear congruences  $ax \equiv b \pmod{p}$  when  $p$  is prime and  $p \nmid a$ : multiply both sides by  $a^{p-2}$  to obtain the solution.

**§ Corollary 2.15 (Fresh form of Fermat)** Let  $p$  be prime and  $a \in \mathbb{Z}$ . Then

$$a^p \equiv a \pmod{p}.$$

*Proof.* If  $p \mid a$  the congruence is trivial. If  $p \nmid a$ , multiply Fermat's Little Theorem  $a^{p-1} \equiv 1 \pmod{p}$  by  $a$  to get  $a^p \equiv a \pmod{p}$ .  $\square$

**§ Corollary 2.16 (case  $a = 2$ )** For every prime  $p$ ,

$$2^p \equiv 2 \pmod{p}.$$

*Proof.* Take  $a = 2$  in Corollary Corollary 2.15 (Fresh form of Fermat).  $\square$

**§ Question (converse for base 2)** Is the converse of Corollary Corollary 2.16 (case  $a = 2$ ) true? That is, if  $n > 1$  and  $2^n \equiv 2 \pmod{n}$ , must  $n$  be prime?

**§ Answer and counterexample (341)** No. Example:  $n = 341 = 11 \cdot 31$  is composite but satisfies  $2^{341} \equiv 2 \pmod{341}$ . Since  $(11, 31) = 1$ , it suffices to check modulo 11 and 31.

By Fermat's Little Theorem,

$$2^{10} \equiv 1 \pmod{11} \Rightarrow 2^{341} \equiv (2^{10})^{34} \cdot 2 \equiv 1^{34} \cdot 2 \equiv 2 \pmod{11},$$

and

$$2^{30} \equiv 1 \pmod{31} \Rightarrow 2^{341} \equiv (2^{30})^{11} \cdot 2^{11} \equiv 1^{11} \cdot 2^{11} \pmod{31}.$$

Since  $2^5 \equiv -1 \pmod{31}$  (or compute directly), we have  $2^{11} \equiv 2 \pmod{31}$ , so  $2^{341} \equiv 2 \pmod{31}$ . Hence  $2^{341} \equiv 2 \pmod{341}$  although 341 is composite.

**§ Definition (pseudoprime to base 2)** A composite integer  $n > 1$  is called a (*base-2*) *pseudoprime* if

$$2^n \equiv 2 \pmod{n}.$$

**§ Remark (pseudoprimes)** 341 is the smallest base-2 pseudoprime. There are infinitely many pseudoprimes and they arise for different bases; pseudoprimes show that the naive converse of Fermat's Little Theorem is false.

**§ Definition (Euler's phi-function)** For  $n \in \mathbb{Z}_{>0}$  the Euler totient function  $\phi(n)$  is the number of integers  $1 \leq x \leq n$

with  $\gcd(x, n) = 1$ ; equivalently,

$$\phi(n) := \#\{x \in \{1, 2, \dots, n\} : \gcd(x, n) = 1\}.$$

**§ Examples (values of  $\phi$ )**

$$\phi(12) = 4 \quad (\text{the classes } 1, 5, 7, 11), \quad \phi(14) = 6 \quad (\text{the classes } 1, 3, 5, 9, 11, 13),$$

and for a prime  $p$ ,  $\phi(p) = p - 1$  (classes  $1, 2, \dots, p - 1$ ).

**§ Theorem 2.17 (Euler's Theorem)** Let  $m > 0$  and  $a \in \mathbb{Z}$  with  $\gcd(a, m) = 1$ . Then

$$a^{\phi(m)} \equiv 1 \pmod{m}.$$

*Proof.* Let  $r_1, r_2, \dots, r_{\phi(m)}$  be the positive integers  $\leq m$  that are relatively prime to  $m$ . Since  $\gcd(a, m) = 1$ , multiplication by  $a$  permutes this reduced residue system: the numbers  $ar_1, \dots, ar_{\phi(m)}$  are all distinct modulo  $m$  and each is relatively prime to  $m$  (see Lemma 1.5 and Lemma 1.14 referenced earlier). Hence the set  $\{ar_i \bmod m\}$  is the same set of residues as  $\{r_i\}$ . Multiplying all residues yields

$$a^{\phi(m)} \prod_{i=1}^{\phi(m)} r_i \equiv \prod_{i=1}^{\phi(m)} r_i \pmod{m}.$$

Since  $\gcd(\prod_i r_i, m) = 1$ , we may cancel the product to obtain  $a^{\phi(m)} \equiv 1 \pmod{m}$ . □

**§ Remark (special case)** Fermat's Little Theorem is the special case  $m = p$  prime, since  $\phi(p) = p - 1$ .

**§ Definition (reduced residue system)** A set of  $\phi(m)$  integers each relatively prime to  $m$  and pairwise incongruent modulo  $m$  is called a *reduced residue system* modulo  $m$ .

**§ Example (reduced residue systems)**  $\{1, 5, 7, 11\}$  is a reduced residue system modulo 12. If  $a$  is coprime to  $m$ , then  $\{ar_1, \dots, ar_{\phi(m)}\}$  is also a reduced residue system modulo  $m$  (this was the idea used in the proof of Euler's Theorem).

**§ Corollary 2.19 (inverse via Euler)** Let  $m > 0$  and  $a \in \mathbb{Z}$  with  $\gcd(a, m) = 1$ . Then  $a^{\phi(m)-1}$  is the multiplicative inverse of  $a$  modulo  $m$ .

*Proof.* From  $a^{\phi(m)} \equiv 1 \pmod{m}$  multiply by  $a^{-1}$  (or rewrite as  $a \cdot a^{\phi(m)-1} \equiv 1$ ) to see  $a^{\phi(m)-1}$  is an inverse of  $a$  modulo  $m$ . □

**§ Remark (use of Corollary Corollary 2.19 (inverse via Euler))** Corollary Corollary 2.19 (inverse via Euler) helps solve linear congruences  $ax \equiv b \pmod{m}$  when  $\gcd(a, m) = 1$ : multiply both sides by  $a^{\phi(m)-1}$  to obtain the solution.

**§ Definition (arithmetic function)** An *arithmetic function* is a function  $f : \mathbb{Z}_{>0} \rightarrow \mathbb{C}$  (or  $\mathbb{R}$ , or  $\mathbb{Z}$ ). Examples: the Euler  $\phi$ -function, the divisor-counting function  $d(n)$ , the sum-of-divisors function  $\sigma(n)$ , etc.

**§ Definition (multiplicative vs completely multiplicative)** An arithmetic function  $f$  is *multiplicative* if  $f(mn) = f(m)f(n)$  whenever  $\gcd(m, n) = 1$ . It is *completely multiplicative* if  $f(mn) = f(m)f(n)$  for all positive integers  $m, n$  (no coprimality required).

**§ Proposition (values determined by prime powers)** If  $f$  is multiplicative and

$$n = p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}$$

is the prime factorization of  $n$ , then

$$f(n) = \prod_{i=1}^r f(p_i^{a_i}).$$

If  $f$  is completely multiplicative, then

$$f(n) = \prod_{i=1}^r f(p_i)^{a_i}.$$

*Proof.* Write  $n$  as a product of relatively prime prime-power factors and apply multiplicativity repeatedly. □

**§ Remark (implication)** Thus knowing a multiplicative arithmetic function on prime powers (or a completely multiplicative function on primes) determines it everywhere.